

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		High frequency / high energy / short wavelength (1) electromagnetic wave (1)	2			2		
		(ii)		Easily pass through the body / highly penetrative(1) with low ionising power (1)	2			2		
		(iii)		Thallium–(201) (1) because it will <u>decay quicker</u> than the others / won't be exposed to radiation for too long(1) Do not accept shortest half life		1	1	2		
	(b)			$1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \frac{1}{16} \rightarrow \frac{1}{32}$ (1) = 5 half lives (1) time = 5 (ecf) × 17 (ecf) (1) ans = 85 days (1) only allow ecf if first line arrives at an answer other than 5 and is shown		4		4	3	
				Question 5 total	4	5	1	10	3	0